onecourse adaptive mode

In order to understand how units are selected for each adaptive session we should first review how the short diagnostic test works.

Diagnostics consists of a series of questions starting from very easy ones (like touch an image that matches said word) and then adjusting until a correct level for the user is achieved. There are 15 levels for literacy and 16 for numeracy. The current adaptive diagnostic starts with level 1 question, if succeeded asks level 8 question and if failed assigns level 1 to the user. From level 8 question success will result in level 12 question, while failure will lead to level 5 question and so on. The user has to answer 2 questions of the same to advance to the next one. Two incorrect answers will either result in lower level questions or a lesson being selected. Sometimes the question level won't change but instead the type will, for instance a level 10 word recognition question could lead to level 10 reading question.

See [example sheet] with following tabs:

- **Questions_literacy/numeracy** – lists all the possible questions that can be asked in diagnostics, they are further explained below. Same type of question can be asked on different levels. For instance a low level word question will include simple words, while a high level one will consists of longer more complicated words.
- **Lessons_group_literacy/numeracy** – groups of lessons that can be assigned in diagnostics
- **Lessons** – lists all the lessons, what lesson group they are assigned to as well as what parameters they cover. There's also categorisation that we use for teacher mode(topic and subtopic) where each lesson can be assigned to up to 3 categories
- **Graph_literacy/numeracy** – a form of binary tree describing how we progress through diagnostics. We start with a question in the first row. Each question will either lead to a next question ("question wrong" and "question correct" columns) or to lesson ("lesson correct" and "lesson wrong" column)
- **Units** – lists all the units with lessons and levels they are assigned to, please note that when I mention playing a lesson I mean playing all units with a selected lesson id(usually between 2 and 4 units). This does not apply for 2 lesson ids: lesson.story and lesson.filler for which we only play a single unit at a time
- **Parameters** – lists parameters used for literacy diagnostics and units, they have a level assigned themselves which allows us to select right parameters for correct level of diagnostic question

Let's first cover the analytics data you can receive for each diagnostics question. An example will look something like this:

JavaScript


```javascript
{"peerId":"f52d745fcf419762","sequenceNumber":208,"payload":{"duration":2,"data":{"answer":"is_o","options":"is_m,is_k,is_d,is_o","status":"correct","target":"is_o","id":"L2_LSQ","topic":"literacy","session_id":5},"timestamp":1696921888635,"session_id":5,"_type":"AD_QUESTION"}}
```

Where:

- **`id`** – the identification of question, prefixed by a level in pattern “LX_ID” where X is the level and ID is the id, you can find the question listed in the excel file
- **`options`** – what options are displayed on the screen, usually ids of words that can be matched with our dictionary
- **`target`** – what is the correct answer
- **`answer`** – what choice has user selected
- **`topic`** – literacy or numeracy
- **`status`** – correct or wrong
- **`extra`** – this will appear for some questions, for instance an id of question asked for word problems

You also receive data for entire diagnostic under **`DIAGNOSTIC`** payload, like this:

JavaScript

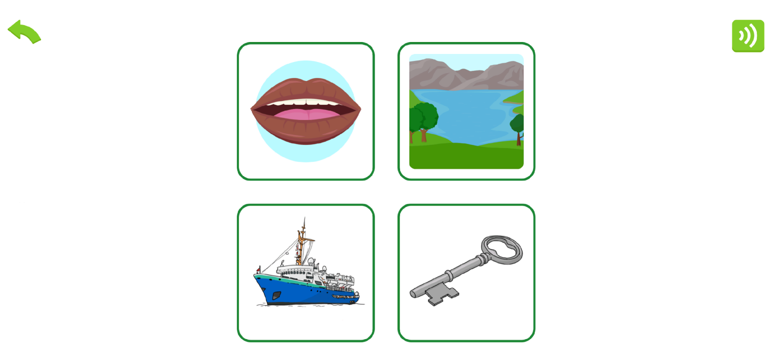
```javascript
{"peerId":"f52d745fcf419762","sequenceNumber":214,"payload":{"duration":129,"data":{"unit_topic":"diagnostic_literacy","unit_key":"0002.OC_ADIntro","type":"diagnostic","literacy":3,"session_id":5},"timestamp":1696921926127,"session_id":5,"_type":"DIAGNOSTIC"}}
```


Where:

- **`unit_topic`** - either “diagnostic_literacy” or “diagnostic_numeracy”
- **`literacy`** – what literacy level achieved at the end of diagnostic(if this is “diagnostic_literacy” topic)
- **`numeracy`** – what numeracy level achieved at the end of diagnostic(if this is “diagnostic_numeracy” topic)

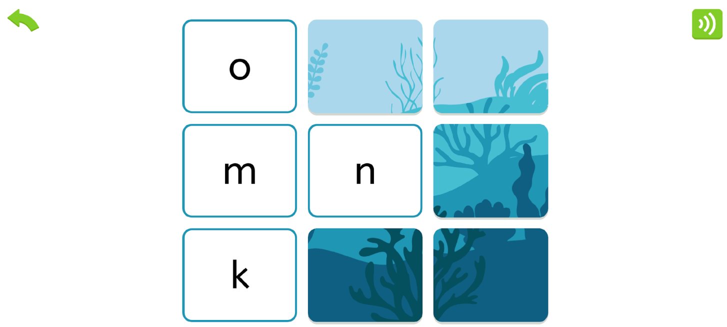
Below you can find list of all questions and levels, please note the level span might vary depending on language, also some questions might not appear for certain languages (e.g. English doesn’t have syllables), These levels are for the Chichewa language:

1. Literacy:
   a. Vocabulary Questions (VQ):
      - Task: pick image that matches word audio
      - Level: 1



b. Letter Sound Question (LSQ):

- Task: pick correct letter for the sound
- Level: 2-5

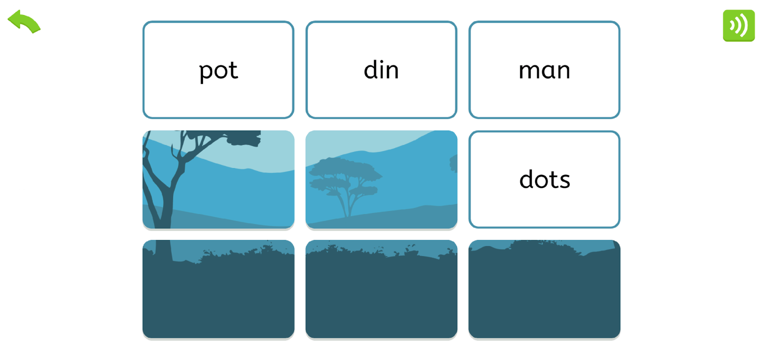


c. Syllable Question (SYQ):

- Task: pick syllable for sound, basically the same style as LSQ above
- Levels: 3-7

d. Word Question (WQ):

- Task: pick word for sound; for higher levels longer and more difficult words will be selected
-
- Levels: 3-15



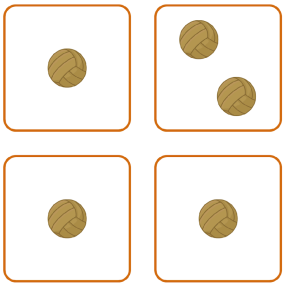
e. Reading Comprehension Question (RCQ):

- Task: pick a missing word for a phrase or sentence
- Levels: 7-15

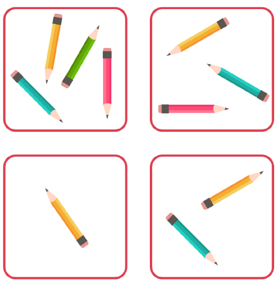
a cat with a ________

on	hum
hat	ill

2. Numeracy:
    a. Sorting Question (SQ):
        - Task: select odd one out
        - Level: 1



    b. Number of Objects Question (NOQ):
        - Task: select image with the correct count of objects
        - Level: 2-3

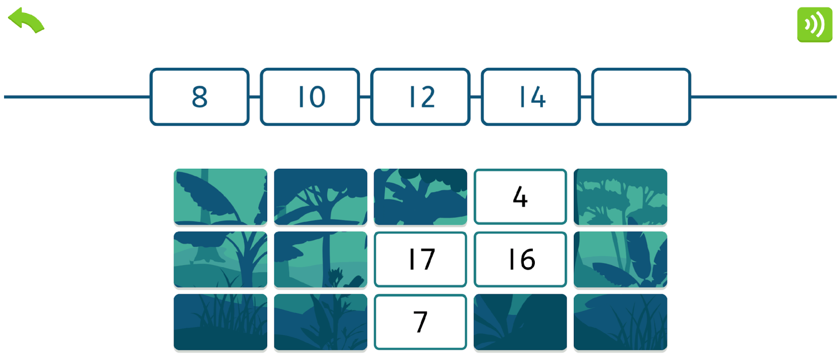


    c. Number Identification Question (NIQ):
        - Task: find a number for given audio; for higher levels bigger numbers will selected(i.e. for level 13 range is 300-999)

- Level: 2-13

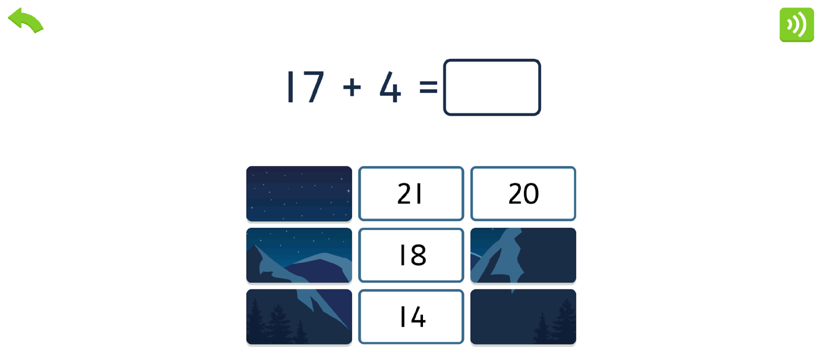
d. Missing Number Question (MNQ):

- Task: find a missing number in a sequence, for higher levels the numbers are higher and the distribution is more complicated(for instance in level 9 numbers are in pattern of multiples of 5)
- Level: 5-12



e. Addition Questions (AEQ):

- Task: solve addition equation
- Level: 6-15



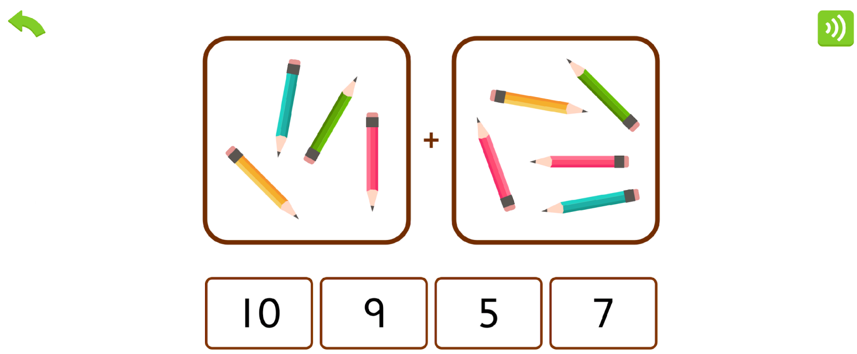
f. Subtraction Question (SEQ):

- Task: solve subtraction equation

- Level: 6-15

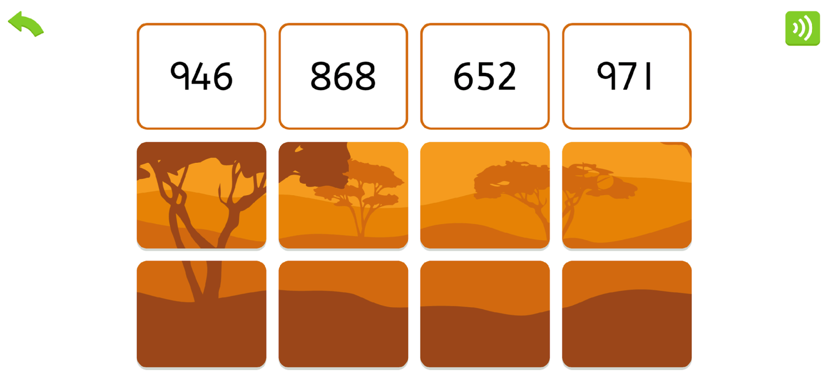
g. Adding Objects Question (AOQ):

- Task: pick a correct answer for equation represented in objects
- Level: 6



h. Number Differentiation Question (NDQ):

- Task: find the biggest number
- Level: 14



i. Word Problems Question (WPQ):

- Task: pick a correct answer to a spoken work problem
- Level: 15

There are 3 main data points we derive from diagnostics in order to select correct units for the user:

- Achieved level
- What was the last question answered(correct or incorrect), we are interested in lesson group that user has landed on the diagnostic graph
- What are the correct answers the user has failed to select

For instance we can get information like this: user had achieved level 2 of literacy, failed on question L2_LSQ(which, according to graph is connected to lesson group L2_LSL) on letters is_o and is_i (which are our dictionary ids for letters o and i).

Based on this information we can start selecting the right lesson to present to user. There are 4 types of lesson we are try to find, each assigned with different weight:

- Weight 50 – Lesson belonging to the selected lesson group, achieved level and matching one of the parameters user failed at
- Weight 20 – Lesson belonging to selected lesson group and achieved level, but the parameters are different to the ones failed by the user
- Weight 15 – Lesson matching the achieved level of different lesson group to the selected one, but matching one of the parameters user has failed at
- Weight 15 - Lesson matching the achieved level of different lesson group and different parameters

The system will try to find a random lesson that matches each case listed above and then out of those 4 select one randomly but with lesson of higher weight having a higher chance being selected.

For example above(level 2, lesson group L2_LSL, parameters is_i and is_o) the list could look like this:

- Weight 50 - lesson.r11.1 (Meet i)
- Weight 20 - lesson.r8.1 (Meet e)
- Weight 15 - lesson.r17.1 (Practice a, e, i, o, u)
- Weight 15 - lesson.r7.1 (Practice a)

The weights attached to each lesson sum up to 100, so you could assume there’s 50% chance (50 out of 100) to select first lesson in the list. However if the system won’t find a lesson to match certain options the distribution will be different. For instance if there's no weight 20 lesson, the chance of weight 50 lesson will now be 62.5% (50 out of 80).

This sums up how a lesson is selected for session. There are few more bits to mention

- User will get both literacy and numeracy diagnostic during a session(unless app is set to only one topic). The order is random
- In sessions that include literacy diagnostics a random book for selected level and with lesson id lesson.story will be selected
- Different session durations will have different amount of units, for BEFIT 50 minutes session we aim for 10 learning units, this count excludes: diagnostics, intro unit(how to use touch screen) and the book mentioned in point above. If the 2 lessons were not enough to fill 10 unit quota the session will end with random units according to weights below(example for default style session):
    - Weight 40 – a single literacy unit with lesson id lesson.filler that matches the achieved level
    - Weight 30 - a single numeracy unit lesson id lesson.filler that matches the achieved level
    - Weight 20 - a literacy lesson that matches the achieved level, that doesn't have more units than remaining count of units(we can't fit a 4 unit lesson if we only have 2 units remaining to achieve 10 unit quota)
    - Weight 10 - same as above but numeracy lesson

The system will keep selecting units according to the above until we have 10 learning units in total.

Below is a list of the kind of data individual units return:

```JavaScript
{"peerId":"7c030bb5e95aa6b9","sequenceNumber":11085,"time":1704797996506,"payload":{"timestamp":1704797901000,"duration":95,"session_uuid":"db65acf4-5515-4851-a739-529456c0e67c","data":{"unit_topic":"literacy","score_correct":5,"status":"completed","unit_key":"2429.OC_Mwyh1","report_type":"study","type":"study","score_wrong":1},"_type":"UNIT"}}
```

Where:

- **`unit_key`** – id of the unit, should much the attached excel
- **`unit_topic`** – literacy, numeracy or general. Very few units have general topic- mostly the "how to use touch screen" unit and some playzone units
- **`status`** – how the unit was finished, options are:
    - "completed" – unit was finished successfully

- “user_closed” – user pressed back button, for instance quit the library book before finishing it
  - “timeout” – the unit has timed out due to inactivity, the usual timeout is 900 seconds for standard units
  - “session_closed” – the unit was closed together with the session, this can happen if user locks the screen for 5 min or more, or after 10 sec pressed the new session button in top left
  - “battery_low” – the unit was closed together with session because device battery was low
  - “failure” – the unit was closed due to crash/bug
- **`type`** – what type of unit is this, options are:
  - “study” – standard study unit
  - “fun” - a fun warm up unit
  - “diagnostics” – diagnostic unit
  - “community” – a replay of study unit in playzone(via star button)
  - “play_zone” – a play zone unit
  - “library” - a library unit
  - “play_zone_asset” – a replay of play zone creation in play zone(bottom bar)
  - “teacher” – a unit played in teacher mode
- **`report_type`** – this is shorhand type information to match the data in USB reports. Some types from list above are merged into one:
  - “study” – encapsulates both “study” and “teacher”
  - “playzone” – encapsulates both “playzone” and “community”
  - “library” – just “library” units
  - “diagnostics” – just “diagnostics”
- **`duration`** – how long did it take to finish the unit in seconds
- **`score_correct`** - how many correct actions has the user performed, this will usually equal to the amount of questions in the unit, however in some units failure will lead to extra questions which means the count will be higher
- **`score_wrong`** – how many failures has the user performed, repeat failures in one question count multiple times